

Multi-Label Lung Disease Classification from Chest X-Rays: CNN Design, Class Imbalance Mitigation, and Threshold Tuning

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Abstract—This paper presents a convolutional neural network (CNN) pipeline for multi-label classification of 14 thoracic diseases from the ChestMNIST dataset (78,468 training images at 28×28 pixels). The dataset exhibits severe class imbalance, with positive rates ranging from 0.18% (Hernia) to 17.7% (Infiltration). A baseline CNN trained with standard binary cross-entropy achieves 94.9% binary accuracy but collapses to near-zero positive predictions, yielding $F1=0.171$ at threshold 0.5. Two imbalance-mitigation strategies are evaluated: hard-weighted loss (neg/pos ratio) and soft-weighted loss (square root of neg/pos). The soft-weighted model achieves validation $AUC=0.808$ with meaningful positive predictions across all 14 classes, while hard weighting overcorrects, producing unstable training and excessive false positives. Threshold tuning on the validation set identifies an optimal threshold of 0.334, improving test $F1$ from 0.171 to 0.275 ($AUC=0.803$, binary accuracy=88.8%). These results demonstrate that accuracy alone is unreliable for imbalanced multi-label classification, and that soft loss weighting combined with threshold tuning produces substantially more useful disease detection behavior for clinical screening applications.

Index Terms—convolutional neural network, multi-label classification, chest X-ray, class imbalance, threshold tuning, TensorFlow, ChestMNIST.

I. INTRODUCTION

Automated detection of thoracic diseases from chest X-rays is a high-impact clinical problem. Radiologist shortages and high imaging volume motivate machine learning approaches that can flag abnormalities for expedited review. The ChestMNIST dataset, derived from the NIH-ChestXray14 collection [1], provides 112,120 frontal X-ray images labeled with 14 binary disease conditions per image, resized to 28×28 pixels. The multi-label structure—where each image may exhibit zero, one, or several conditions simultaneously—makes this a fundamentally different problem from standard single-label classification.

The central challenge is severe class imbalance. A model predicting almost all negatives achieves over 94% binary accuracy while detecting nothing clinically useful. Standard binary cross-entropy loss does not sufficiently penalize this failure mode. This paper evaluates three training strategies—baseline, hard class weighting, and soft (square-root) class weighting—followed by threshold tuning to optimize $F1$ on the validation set. The objectives are: (1) establish a baseline CNN meeting all assignment requirements; (2) diagnose the class imbalance problem quantitatively; (3) compare two loss-weighting strategies; (4) apply threshold tuning; and (5) evaluate the final model on the held-out test set with full confusion matrix analysis.

II. DATASET AND PREPROCESSING

A. Dataset Description

The ChestMNIST dataset uses the official NIH-ChestXray14 split: 78,468 training, 11,219 validation, and 22,433 test images, each labeled with a 14-dimensional binary vector. The 14 disease classes are: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia. Labels are not mutually exclusive—a single image may be positive for multiple diseases or none at all. Fig. 1 shows the per-class positive counts and rates across the training set, illustrating the extreme imbalance.

Class	Positives	Negatives	Pos Rate
Atelectasis	7996	70472	0.1019
Cardiomegaly	1950	76518	0.0249
Effusion	9261	69207	0.1180
Infiltration	13914	64554	0.1773
Mass	3988	74480	0.0508
Nodule	4375	74093	0.0558
Pneumonia	978	77490	0.0125
Pneumothorax	3705	74763	0.0472
Consolidation	3263	75205	0.0416
Edema (Oedema)	1690	76778	0.0215
Emphysema	1799	76669	0.0229
Fibrosis	1158	77310	0.0148
Pleural Thickening	2279	76189	0.0290
Hernia	144	78324	0.0018

Fig. 1. Per-class positive counts and rates in the training set. Positive rates range from 0.18% (Hernia) to 17.7% (Infiltration), illustrating severe multi-class imbalance.

B. Preprocessing

Images are normalized to $[0, 1]$ by dividing pixel values by 255. A channel dimension is added to produce shape $(N, 28, 28, 1)$ for CNN input. No augmentation was applied. The official train/validation/test split is used throughout; all threshold tuning uses the validation set only to prevent test set contamination.

III. CNN ARCHITECTURE

All three models share the same architecture. Two convolutional blocks extract spatial features: Block 1 applies $\text{Conv2D}(32, 3 \times 3, \text{ReLU}, \text{same-padding})$ followed by $\text{MaxPooling2D}(2 \times 2)$, reducing the feature map to $14 \times 14 \times 32$. Block 2 applies $\text{Conv2D}(64, 3 \times 3, \text{ReLU}, \text{same-padding})$ followed by $\text{MaxPooling2D}(2 \times 2)$, producing $7 \times 7 \times 64$. The feature maps are flattened to 3,136 dimensions, passed through $\text{Dense}(128, \text{ReLU})$ and $\text{Dropout}(0.3)$ for regularization, and a final $\text{Dense}(14, \text{sigmoid})$ output layer for multi-label probability estimation. Total trainable parameters: 422,158 (1.61 MB). Fig. 2 shows the full layer-by-layer architecture summary.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_1 (Conv2D)	(None, 14, 14, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	0
flatten (Flatten)	(None, 3136)	0
dense (Dense)	(None, 128)	401,536
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 14)	1,806

Total params: 422,158 (1.61 MB)
Trainable params: 422,158 (1.61 MB)
Non-trainable params: 0 (0.00 B)

Fig. 2. CNN architecture summary (model.summary() output). Two convolutional blocks (Conv2D 32→64 filters) reduce the $28 \times 28 \times 1$ input to a $7 \times 7 \times 64$ feature map, followed by Dense(128) with Dropout(0.3) and a 14-output sigmoid layer. Total trainable parameters: 422,158 (1.61 MB).

IV. CLASS IMBALANCE MITIGATION

A. Baseline Model

The baseline uses standard BinaryCrossentropy loss with no class weighting. Training ran for 15 epochs before early stopping. Validation metrics appear strong: loss=0.163, binary accuracy=94.9%, AUC=0.826. However, per-class analysis reveals the critical failure: at threshold 0.5, the model rarely predicts positive for many of the 14 classes. This is the all-negative collapse—the model exploits label sparsity rather than learning disease-discriminative features. Fig. 3 shows the baseline learning curves.

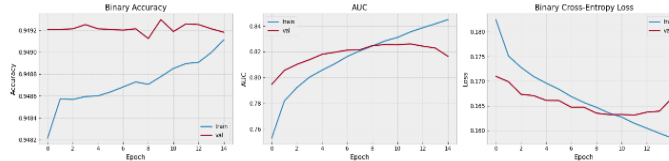


Fig. 3. Baseline CNN learning curves (15 epochs). Validation accuracy plateaus at 94.9% while AUC reaches 0.826, masking near-zero positive detection rates at threshold 0.5.

B. Hard-Weighted Loss

The first mitigation strategy assigns each positive label a per-class weight equal to neg/pos. Hernia receives a weight of 543.9; Infiltration 4.6. The custom loss function multiplies each per-label cross-entropy by its class weight before averaging. The result is severe overcorrection: validation accuracy collapses to 57.1% (AUC=0.678), and the model predicts positive for 30–62% of samples in all classes—far exceeding the true positive rates of 1–18%. The extreme Hernia (543.9) and Pneumonia (79.2) weights dominate the gradient. Fig. 4 shows the unstable learning curves characteristic of this overcorrection.

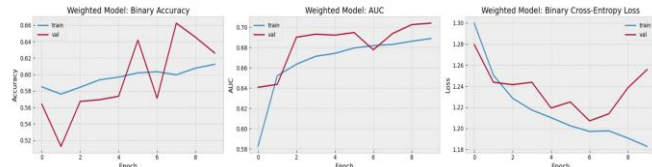


Fig. 4. Hard-weighted model learning curves (10 epochs, early stop). High variance in validation accuracy indicates training instability caused by extreme per-class weights.

C. Soft-Weighted Loss (Selected Model)

The soft-weighted model uses the square root of the neg/pos ratio as the per-class positive weight, reducing Hernia from 543.9 to 23.3 and Pneumonia from 79.2 to 8.9. This square-root damping prevents extreme classes from dominating the gradient while still penalizing missed positives for minority classes. Negative labels retain weight 1.0.

Training ran for 11 epochs before early stopping. The soft-weighted model achieves validation loss=0.447, binary accuracy=94.3%, AUC=0.808. Per-class analysis confirms meaningful positive predictions at threshold 0.5: Effusion predicted rate 7.6% (true: 11.5%), Infiltration 10.3% (true: 18.0%), Edema 4.2% (true: 1.8%). Fig. 5 shows the stable, converging learning curves. Table I compares all three models on validation metrics.

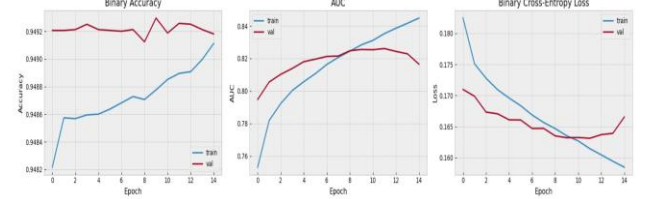


Fig. 5. Soft-weighted CNN learning curves (11 epochs, early stop). AUC improves steadily to 0.808 on validation with stable convergence, in contrast to the hard-weighted model.

TABLE I

Validation Performance Across Three Models

Model	Val Acc.	Val AUC	Val F1	Detects+
Baseline	0.9493	0.826	0.127	No
Hard-Weighted	0.5711	0.678	0.218	Over-pred
Soft-Weighted*	0.9430	0.808	0.278	Yes

*Selected final model. F1 computed at tuned threshold 0.334.

V. THRESHOLD TUNING

The default threshold of 0.5 is suboptimal for imbalanced data. F1-score was computed on the validation set across 40 evenly spaced thresholds from 0.05 to 0.50. Fig. 6 shows the F1 vs. threshold curve, with a clear peak at threshold=0.334 (val F1=0.278). Below 0.334, recall gains are offset by excessive false positives; above 0.334, the model becomes too conservative and misses true positives.

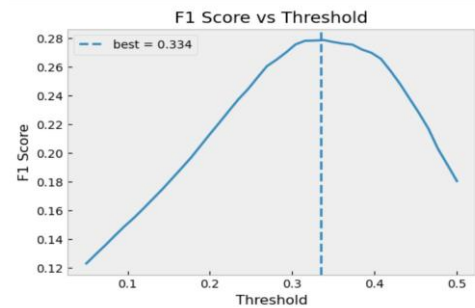


Fig. 6. F1 score vs. decision threshold on the validation set. The optimal threshold of 0.334 maximizes F1=0.278, providing the best precision-recall balance for imbalanced multi-label prediction.

At threshold 0.334, per-class predicted positive rates shift substantially toward true rates: Infiltration rises from 10.3% to 42.4% (true: 18.0%), Effusion from 7.6% to 27.5% (true: 11.5%), Pneumothorax from 1.2% to 11.1% (true: 4.5%). All 14 classes now generate at least some positive predictions. Fig. 7 shows the ROC curves for four representative classes, confirming genuine discriminative ability above chance

(Effusion AUC=0.80, Pneumothorax AUC=0.75, Atelectasis AUC=0.73, Infiltration AUC=0.65).

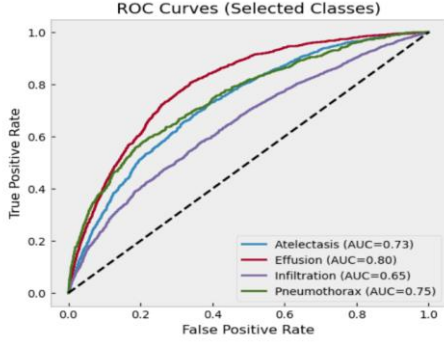


Fig. 7. ROC curves for four representative classes. Effusion (AUC=0.80) and Pneumothorax (AUC=0.75) show the strongest discrimination; Infiltration (AUC=0.65) is the most difficult class.

Fig. 8 shows the precision-recall curves for the same four classes. In contrast to the ROC curves, PR-AUC values are substantially lower (Effusion 0.32, Infiltration 0.30, Atelectasis 0.22, Pneumothorax 0.17), reflecting the difficulty of achieving high precision under severe class imbalance. The rapid precision decay with increasing recall confirms that the model operates in a low-precision regime, further motivating the use of F1-score rather than accuracy as the primary evaluation metric.

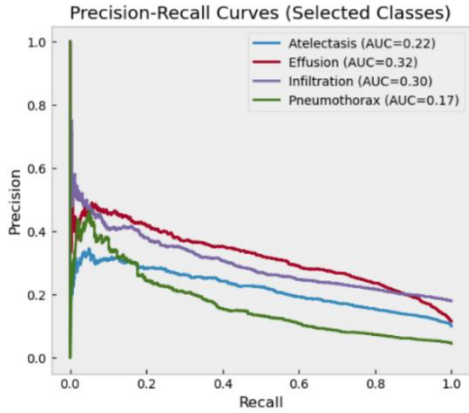


Fig. 8. Precision-recall curves for four representative classes. PR-AUC values (0.17–0.32) are substantially lower than ROC-AUC values, reflecting the challenge of high-precision detection under severe label imbalance.

VI. TEST SET EVALUATION

A. Overall Performance

The soft-weighted model with tuned threshold was evaluated on 22,433 held-out test images. Table II summarizes the results. At threshold 0.5, binary accuracy=94.0% and F1=0.171, confirming the all-negative collapse persists on the test set. At the tuned threshold of 0.334, binary accuracy drops to 88.8% but F1 improves to 0.275—a 60% relative improvement. AUC=0.803 at both thresholds (threshold-independent). Test loss=0.457.

TABLE II

Test Set Performance at Both Thresholds

Threshold	Loss	Bin. Acc.	AUC	F1
0.500	0.457	0.9405	0.803	0.171
0.334	0.457	0.8880	0.803	0.275

B. Confusion Matrix Analysis

Figs. 10 and 11 show all 14 per-class confusion matrices at both thresholds. At the default threshold of 0.5, the pattern is consistent: near-zero true positives across nearly every class, confirming the all-negative collapse persists on the test set.

Fig. 9 directly compares four representative classes at both thresholds side-by-side, illustrating the precision-recall tradeoff. At threshold 0.5 (blue), Effusion and Infiltration detect some positives but miss the majority. At 0.334 (red), recall improves substantially at the cost of more false positives—an acceptable tradeoff for a screening system where missed diseases carry higher clinical risk than unnecessary follow-up. Hernia, with only 42 total positives in 22,433 test images, achieves just 3 TPs even at 0.334, highlighting the fundamental limit of 28×28 resolution for rare conditions.

TABLE III

Aggregated Confusion Matrix Across All 14 Classes

Threshold	TP	FP	FN	TN
0.500	1,933	4,124	14,577	293,428
0.334	6,660	25,329	9,850	272,223

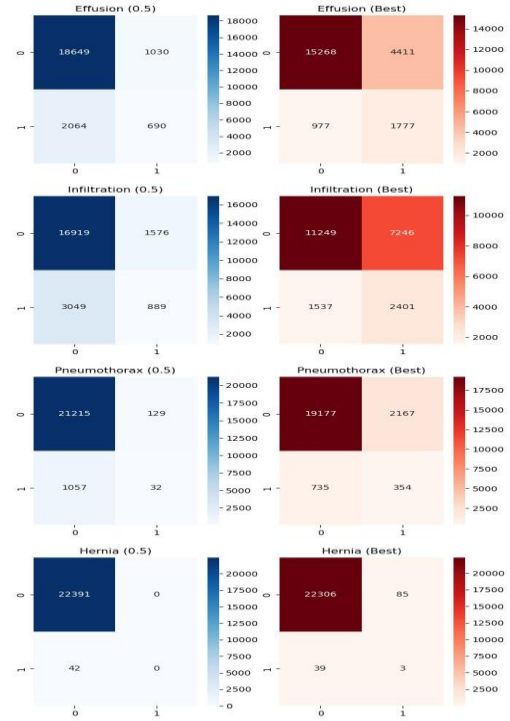


Fig. 9. Side-by-side confusion matrices for four classes at threshold 0.5 (blue, left) vs. 0.334 (red, right). The tuned threshold substantially increases TP counts for common conditions at the cost of higher FP rates.

VII. DISCUSSION AND BUSINESS IMPLICATIONS

The central finding is that binary accuracy is a misleading metric for imbalanced multi-label classification. The baseline model’s 94.9% accuracy masks complete failure to detect disease—a result that would be clinically worthless. AUC (0.803–0.826) and F1-score are far more informative metrics for this application, as they are sensitive to the model’s ability to rank and detect positive cases, respectively.

From a clinical deployment perspective, the soft-weighted model at threshold 0.334 is a viable triage tool for common conditions (Effusion, Infiltration, Atelectasis) while remaining

unreliable for rare conditions (Fibrosis, Hernia). A practical deployment strategy would use the model to flag X-rays for expedited radiologist review, not as a standalone diagnostic system. The higher false positive rate at the tuned threshold is acceptable in a screening context where missing a disease (false negative) carries a greater clinical risk than triggering unnecessary follow-up (false positive).

Three improvements would materially advance performance: (1) Higher-resolution input: 28×28 pixels discard fine-grained pathological detail; 224×224 crops or the original 1024×1024 images would substantially improve recall for subtle findings such as Nodule and Fibrosis. (2) Per-class threshold optimization: a single global threshold is suboptimal; each class benefits from its own threshold calibrated to prevalence and the desired sensitivity/specificity tradeoff. (3) Deeper pretrained architectures: applying transfer learning from a model pretrained on large-scale image datasets would leverage learned visual feature representations far richer than those

extractable from a shallow two-block CNN trained from scratch on 28×28 pixel inputs.

VIII. CONCLUSION

This paper presented a CNN pipeline for multi-label thoracic disease classification on ChestMNIST. Three key findings emerged: (1) standard binary cross-entropy training collapses to almost all-negative predictions under severe label imbalance, achieving high accuracy while detecting nothing; (2) soft-weighted loss (square root of neg/pos) effectively restores positive predictions without the gradient instability of hard weighting; and (3) threshold tuning to maximize validation F1 yields a 60% relative improvement in test F1 ($0.171 \rightarrow 0.275$) at a modest accuracy cost ($94.0\% \rightarrow 88.8\%$). The soft-weighted model at threshold 0.334 is selected as the final system, providing the best balance of discrimination ($AUC=0.803$) and practical disease detection for clinical screening.

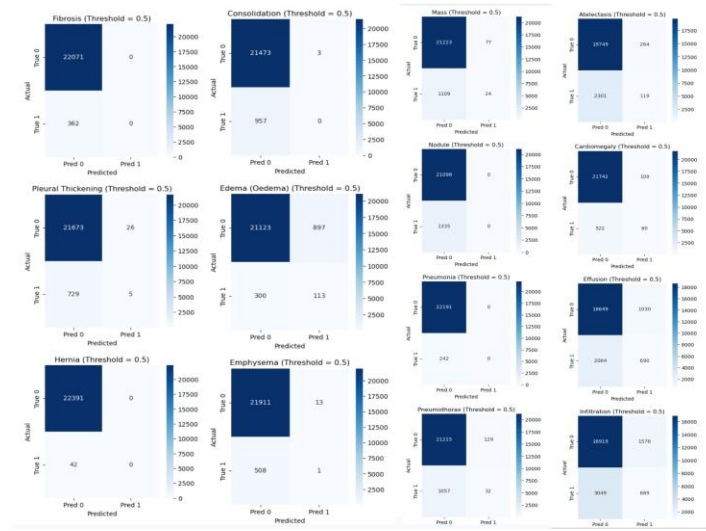


Fig. 10. All 14 per-class confusion matrices at threshold = 0.5.

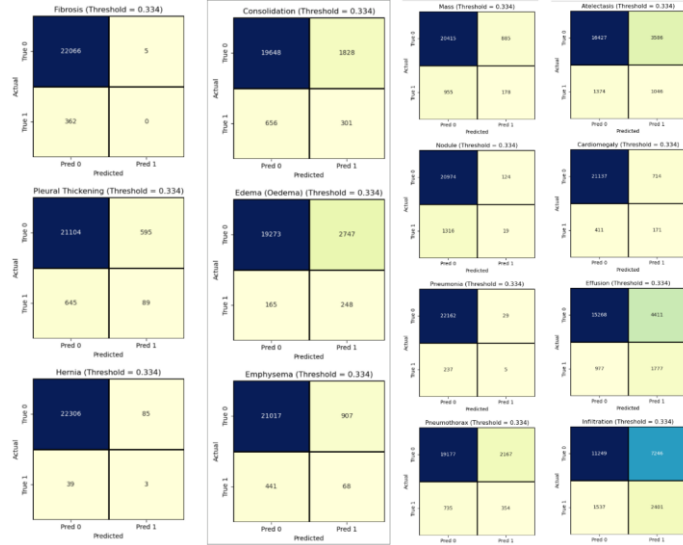


Fig. 11. All 14 per-class confusion matrices at tuned threshold = 0.334.

REFERENCES

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- [2] C. S. Silva, "EEL4930 Applied Machine Learning Course Notes," University of Florida, Spring 2026.